

Sadig Akhund

Data Engineer

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Summary

Data Engineer with 3+ years building large-scale data pipelines, warehouses, and distributed systems. Delivered production-grade ETL infrastructure processing hundreds of millions of records for government and financial clients. Core expertise: **Apache Spark, Kafka, Iceberg, Trino**, cloud platforms, and full-stack data platform development. Author and maintainer of an open-source Python ETL framework.

Experience



KDM Force

Data Engineer (Remote)

Rome, Italy

Jun 2022 – Nov 2025

Stack: Apache Spark, Kafka, Iceberg, Trino, Hive Metastore, PostgreSQL, Redis, MS SQL Server, Dataiku, RHEL

- Built end-to-end data ingestion pipelines from scratch using **Apache Spark** and the Fyrefuse orchestration platform, loading OTA data into a MS SQL Server warehouse for the **Ministry of Tourism of Saudi Arabia**.
- Designed a warehouse schema supporting **80M+ customer reviews** and multi-year listing metadata across **18 business-critical tables**, powering national tourism analytics.
- Integrated **Trino, Hive Metastore, and Apache Iceberg** with object storage, opening up federated analytical queries over historical data without re-ingesting to the warehouse.
- Built a **viewport-aware caching layer** between the application frontend and Trino-backed sources — preloaded pages around the user's scroll window in both directions, eliminating perceived load time on multi-million-row CSV-style tables.
- Selected **PostgreSQL** over Redis as the cache backend after benchmarking: Redis covered only key-value access while users needed sort, filter, aggregation, and full-text search; Postgres delivered all four plus **sub-second write times for 1M records** after switching the ingest path from `INSERT/bulk-insert` to `COPY`.
- Engineered adaptive error handling for unreliable OTA providers (rate-limit backoff, schema drift detection, dead-letter queue), eliminating routine manual intervention.
- Owned ETL workflow governance in **Dataiku** — parameter file management, log auditing, pipeline observability across all ingestion jobs.



Unibank

Data Engineer (Intern)

Baku, Azerbaijan

Oct 2021 – May 2022

Stack: Apache Spark, Apache NiFi, Elasticsearch, Logstash, Kibana, Python, FastAPI, MongoDB, InfluxDB, Snowflake

- Studied and documented the bank's production **push-notification telemetry pipeline** (mobile app → **Apache NiFi** ingestion → **Apache Spark** enrichment → **ELK stack**), surfacing schema gaps and partition hotspots to the platform team.
- Refactored middleware log schema feeding Logstash, cutting per-event payload size and standardizing the format consumed by Spark enrichment jobs — reduced storage cost and improved Kibana query latency.
- Built a **FastAPI** service that exposed enriched notification metrics to internal stakeholders, replacing manual ad-hoc Elasticsearch queries.
- Benchmarked **InfluxDB, Snowflake, and MongoDB** as candidate stores for time-series and document workloads, producing a feasibility report that informed the bank's analytics roadmap.

Projects



Aquaduct · Declarative Spark engine with LLM-driven self-healing [GitHub](https://github.com/sadigaxund/aquaduct)

Apr 2026 – Present

Acquired Skills: Apache Spark, Apache Airflow, Distributed Computing, LLM Agents, CI/CD, Observability, Open-Source

- Designed and shipped a **declarative pipeline engine** (Parser → Compiler → Executor → Surveyor) replacing imperative PySpark with version-controlled YAML Blueprints. Released **1.0.0 to PyPI** as `aquaduct-core`.
- Built a multi-provider **LLM self-healing agent loop** that diagnoses failed Spark runs from a structured `FailureContext` and emits validated patches via a **10-op PatchSpec grammar**; deterministic guardrails + provenance-aware prompts cut hallucination.

- Shipped **first-class Apache Airflow integration**: drop-in `AqueductOperator` + deferred-pattern sensors; supports human/CI healing modes via structured exit codes. Replaces the imperative-Spark layer underneath existing orchestration.
- Engineered **zero-cost observability**: per-module metrics via `DataFrame.observe()` + `SparkListener`, column-level lineage via `sqlglot`, embedded DuckDB store — no extra Spark actions on the critical path.
- Implemented **pluggable persistence** (DuckDB, Postgres, Redis behind a Store API), multi-cloud secrets dispatch (AWS / GCP / Azure), structured Spark error extraction, and a two-tier benchmark harness for prompt regression detection in CI.

Certifications



Google Data Analytics Professional Certificate · [Verify on Credly](#)

Apr 2022

Education



George Washington University & ADA University

2021 – 2023

M.Sc., Computer Science & Data Analytics



ADA University

2017 – 2021

B.Sc., Computer Engineering

Skills

Data Engineering	Apache Spark, Kafka, Airflow, dbt, Iceberg, Delta Lake, Trino, ETL/ELT
Languages	Python, SQL, Java, C++, Bash
Databases	PostgreSQL, Snowflake, BigQuery, DuckDB, MS SQL Server, MongoDB, Redis
Cloud & DevOps	GCP, AWS, Azure, Docker, Kubernetes, Terraform, GitHub Actions, Linux
LLM & Agents	Anthropic, OpenAI, Ollama, agent loops, prompt engineering, structured output
Backend & Tools	FastAPI, Django, Pydantic, Click, sqlglot, Git, REST APIs
Analytics & BI	Looker, Superset, Grafana, Metabase, Data Modeling
Spoken Languages	English (Proficient) Russian (Advanced) Azerbaijani (Native)